$$C(\mathbf{r}) = \int_{t_n}^{t_f} T(t)\sigma(\mathbf{r}(t))\mathbf{c}(\mathbf{r}(t), \mathbf{d})\, dt,$$

$$T(t) = \exp\left[-\int_{t_n}^{t} \sigma(\mathbf{r}(s))\, ds\right]$$